Phase 1-Configure GoldenGate Replication from Oracle Range to Oracle Hash

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• Setup and configure replication from Range RDS to Hash RDS

• Setup Environment

• Configure GG Replication from Range Partitioning to Hash Partitioning

• Create New GG Hub to setup replication for Hash RDS

```
./create_gg_hub_ha_new.sh psp prod psphpp01 us-west-2 vpc-2 no m4.10xlarge 1000
```

• Create and start Pump to send trails to HASH GG hub

- Login to current production GG hub and run below commands

```
GGSCI (ggpspuwp01.sbg-psp-prod.a.intuit.com) 2> obey ./diroby/create_ppsphp01.oby

GGSCI (ggpspuwp01.sbg-psp-prod.a.intuit.com) 3> add extract ppsphp01, EXTTRAILSOURCE

EXTRACT added.


GGSCI (ggpspuwp01.sbg-psp-prod.a.intuit.com) 4> add rmttrail ./dirdat/pspuwp01/tr, e

RMTTRAIL added.

GGSCI (ggpspuwp01.sbg-psp-prod.a.intuit.com) 5> info all

Program     Status      Group       Lag at Chkpt  Time Since Chkpt
```

```
MANAGER     RUNNING
EXTRACT     RUNNING     EPSPWP01    00:00:02      00:00:00
EXTRACT     RUNNING     EUIPWP01    00:00:03      00:00:03
EXTRACT     ABENDED     PPSPEP01    00:00:00      00:04:40
EXTRACT     RUNNING     PPSPEP05    00:00:03      00:00:08
EXTRACT     STOPPED     PPSPHP01    00:00:00      00:00:05
EXTRACT     RUNNING     PUIPWP01    00:00:03      00:00:09
```

- ## Restore latest snapshot of Range RDS via console.

```
------- ------------------------------

e#, resetlogs_time, prior_resetlogs_change#, prior_resetlogs_time, status, resetlogs_i

  RESETLOGS_CHANGE# RESETLOGS        PRIOR_RESETLOGS_CHANGE# PRIOR_RES STATUS
-------------------- ---------- ------------------------------ ---------- -------- ------
      8388795358333 14-AUG-20                  8388795357611 14-AUG-20 PARENT
      8397003383021 07-APR-23                  8388795358333 14-AUG-20 PARENT
      8397003385952 07-APR-23                  8397003383021 07-APR-23 CURRENT
```

- Increase RDS storage for PITHASH1 to have ~14TB free for export backup & Take full database export in above restored RDS

- ## Create New HASH RDS

```
./create_rds.sh prodx clusterdb psphpp02 us-west-2 vpc-2

--Change Master user password
aws --profile sbg-psp-prod --region us-west-2 rds modify-db-instance --db-instance-i
```

- Initiai Sync
- **Export data from PITHASH1 db**
- create directories in PITHASH1 db & Hash rds

```
EXEC rdsadmin.rdsadmin_util.create_directory(p_directory_name => 'DATA_PUMP_DIR1');
EXEC rdsadmin.rdsadmin_util.create_directory(p_directory_name => 'DATA_PUMP_DIR2');
EXEC rdsadmin.rdsadmin_util.create_directory(p_directory_name => 'DATA_PUMP_DIR3');
select * from DBA_DIRECTORIES where DIRECTORY_NAME like 'DATA_PUMP_DIR%';
```

- create database link in PITHASH1 db

```
drop database link to_target;
create database link to_target connect to intuadmin identified by "xxxxxx" using '(D
select INSTANCE_NAME, HOST_NAME from v$instance@to_target;
```

- export metadata

```
nohup expdp userid=intuadmin/xxxxx@pithash1 directory=DATA_PUMP_DIR dumpfile=exp_psp
```

- export data

```
nohup expdp userid=intuadmin/"xxxx"@pithash1 dumpfile=DATA_PUMP_DIR:exp_data_%U.dmp,

--export dump size : 13.893 TB (~14 TB) (Time Taken: ~11 hrs)
```

- Copy dump files to target Hash db

```
nohup ./pithash1_run_on_target.sh copy_dump_files_to_target_p1.sql &
nohup ./pithash1_run_on_target.sh copy_dump_files_to_target_p2.sql &
nohup ./pithash1_run_on_target.sh copy_dump_files_to_target_p3.sql &
nohup ./pithash1_run_on_target.sh copy_dump_files_to_target_p4.sql &
nohup ./pithash1_run_on_target.sh copy_dump_files_to_target_p5.sql &
nohup ./pithash1_run_on_target.sh copy_dump_files_to_target_p6.sql &
nohup ./pithash1_run_on_target.sh copy_dump_files_to_target_p7.sql &
nohup ./pithash1_run_on_target.sh copy_dump_files_to_target_p8.sql &

--Overall time taken ~8 hrs
```

- **Import data to HASH RDS db**
- Change DB parameter group to have new undo retention as 900 (15 mins)
- Recreate redo logs

```
./run_on_target.sh add_redos.sql

--ensure no errors
grep ORA- add_redos.lst
```

- create tablespaces on Hash RDS

```
nohup ./run_on_target.sh create_tablespaces.sql &

--ensure no errors
grep ORA- create_tablespaces.lst

--Temprarily change TEMP and UNDO tablespace size
alter tablespace temp resize 5T;
alter tablespace UNDO_T1 resize 4T;
```

- Create schema owners

```
./run_on_target.sh create_schema_owners.sql
```

- Create tables with PK

```
./run_sql.sh pspadm_hash_Partitioned_tables_ddl.sql
./run_sql.sh tables.sql
./run_sql.sh create_pk.sql
```

- import data

```
nohup impdp userid=intuadmin/"xxxxxx"@psphpp02 dumpfile=DATA_PUMP_DIR:exp_data_%U.dm
--Time taken ~15 hrs

nohup impdp userid=intuadmin/"xxxxxx"@psphpp02 dumpfile=DATA_PUMP_DIR:exp_data_%U.dm
--Time taken 16 hrs 30 mins

nohup impdp userid=intuadmin/"xxxxxx"@psphpp02 dumpfile=DATA_PUMP_DIR:exp_data_%U.dm
 --Time taken ~11 hrs 45 mins

nohup impdp userid=intuadmin/"xxxxxx"@psphpp02 dumpfile=DATA_PUMP_DIR:exp_data_%U.dm
 --Time taken ~5 hrs 20 mins

nohup impdp userid=intuadmin/"xxxxxx"@psphpp02 dumpfile=DATA_PUMP_DIR:exp_data_%U.dm
 --Time taken ~4 hrs 30 mins

--Ensure No Errors in all import logs
```

- run row count sanity (in case of import failures seen in import log)
- collect stats

```
nohup ./run_sql.sh collect_stats.sql &
--Time taken ~6 hrs 30 mins
```

- cleanup dump files

```
nohup ./run_sql.sh cleanup_exp_dump_files.sql &
```

- Create other Indexes

```
./run_sql.sh create_sec_fk_indexes_1.sql
./run_sql.sh create_sec_fk_indexes_2.sql
./run_sql.sh create_sec_fk_indexes_3.sql
./run_sql.sh create_sec_fk_indexes_4.sql

--Time Taken ~27 hrs
```

- (Optional) Below is optional in case there is any error while index creation

```
CREATE INDEX PSPADM.PSP_PSTUB_PAY_ITEM_FK1 ON PSPADM.PSP_PSTUB_PAY_ITEM (COMPANY_FK,

--got below error for few indexes, so created indexes offline as below(without using
ORA-00604: error occurred at recursive SQL level 1
ORA-01450: maximum key length (3215) exceeded
--offline rebuild
CREATE INDEX PSPADM.PSP_AGENCY_AGENCYIDENC_I1 ON PSPADM.PSP_AGENCY (AGENCY_ID_ENC) p
CREATE INDEX PSPADM.PSP_BANK_ACCOUNT_ENC_I1 ON PSPADM.PSP_BANK_ACCOUNT (ACCOUNT_NUME
CREATE INDEX PSPADM.PSP_COMPANY_FEDTAXIDENC_I1 ON PSPADM.PSP_COMPANY (FED_TAX_ID_ENC

CREATE INDEX PSPADM.PSP_EFTPS_PAYMENT_DETAIL_I3 ON PSPADM.PSP_EFTPS_PAYMENT_DETAIL (
CREATE INDEX PSPADM.PSP_EFTPS_PAYMENT_DETAIL_I4 ON PSPADM.PSP_EFTPS_PAYMENT_DETAIL (
CREATE INDEX PSPADM.PSP_EMPLOYEE_TAXIDENC_I1 ON PSPADM.PSP_EMPLOYEE (TAX_ID_ENC)  pa
CREATE INDEX PSPADM.PSP_ENMT_FEDTAXIDENC_I1 ON PSPADM.PSP_ENTITLEMENT_UNIT (FED_TAX_

CREATE INDEX PSPADM.PSP_FRAUD_ACCOUNT_ENC_I1 ON PSPADM.PSP_FRAUD_BANK_ACCOUNT (ACCOU
```

- disable index parallel degree

```
./run_sql.sh disable_index_parallel.sql &
```

- create FK constraints

```
./run_sql.sh ref_constraints.sql

NOTE:
No FK between PSP_FINTXN_ONHOLDREASON_ASSOC & PSP_FINANCIAL_TRANSACTION
No FK between PSP_PAYCHECK_USAGE_HIST & PSP_PAYCHECK_USAG
```

- Import other objects

```
nohup impdp userid=intuadmin/"xxxxxx"@psphpp02 dumpfile=DATA_PUMP_DIR:exp_pspadm_met

impdp userid=intuadmin/"xxxxxx"@psphpp02 dumpfile=DATA_PUMP_DIR:exp_pspadm_metadata.
```

- create check and not null constraints via generated script from source(production)

```
./run_sql.sh create_notnull_cc.sql
```

```
--Ensure no errors
```

- create users

```
./run_sql.sh create_users.sql

--below should not return any PSPADM objects errors
grep -B 3 ORA-00942 CREATE_USERS.lst|egrep -v '"PSPBOADMIN"."|"APPBUGFIX"."|"MCHOUBE
```

- lock app users

```
alter user pspapp account lock;
alter user pspadm account lock;
```

- check for invalid objects.

```
select owner, object_type, object_name from dba_objects where status <> 'VALID';
```

- Run Sanity check on source and target HASH rds db

```
@sanity_chk.sql
```

- ## Final step for Goldengate replication from Range to Hash RDS
- login to HASH RDS
- Setup supplemental log

```
SELECT SUPPLEMENTAL_LOG_DATA_MIN, FORCE_LOGGING FROM V$DATABASE;
exec rdsadmin.rdsadmin_util.alter_supplemental_logging('ADD');
exec rdsadmin.rdsadmin_util.force_logging();
exec rdsadmin.rdsadmin_util.switch_logfile;
exec rdsadmin.rdsadmin_util.set_configuration('archivelog retention hours',72);
commit;
```

- Grant permission to GGT and GGS

```
grant create session to GGT;
grant DBA to GGT;
exec rdsadmin.rdsadmin_dbms_goldengate_auth.grant_admin_privilege(grantee=>'GGT',gra
grant exempt access policy to GGT;

GRANT SELECT ANY DICTIONARY TO GGS;
host sed '8,25d' $GG_HOME/sequence.sql > $GG_HOME/rds_sequence.sql
@$GG_HOME/rds_sequence.sql ggs
GRANT EXECUTE ON GGS.updateSequence to GGT;
GRANT EXECUTE ON GGS.replicateSequence to GGT;
drop table GGT.CHECKPOINT;
drop table GGT.CHECKPOINT_LOX;
```

- Create credentials and checkpoint.

```
ALTER CREDENTIALSTORE ADD USER ggs@psphpp02 PASSWORD "xxxxx" ALIAS ggsource_psphpp02
ALTER CREDENTIALSTORE ADD USER ggt@psphpp02 PASSWORD "xxxxx" ALIAS ggtarget_psphpp02
obey ./diroby/dblogin_s_psphpp02.oby
obey ./diroby/dblogin_t_psphpp02.oby
add checkpointtable
```

- add schema trandata

```
obey ./diroby/dblogin_s.oby
add schematrandata PSPADM
```

- Setup and start oracle range to oracle hash replicat process

```
obey ./diroby/create_rpspwp01.oby

GGSCI (ggpsphpp01.sbg-psp-prod.a.intuit.com) 2> obey ./diroby/create_rpspwp01.oby

GGSCI (ggpsphpp01.sbg-psp-prod.a.intuit.com) 3> dblogin USERIDALIAS ggtarget

Successfully logged into database.

GGSCI (ggpsphpp01.sbg-psp-prod.a.intuit.com as ggt@PSPHPP01) 4> add replicat rpspwp(

REPLICAT (Integrated) added.

GGSCI (ggpsphpp01.sbg-psp-prod.a.intuit.com as ggt@PSPHPP01) 11> register replicat r

2023-03-31 06:29:18  INFO    OGG-02528  REPLICAT RPSPWP01 successfully registered wi
```

- Get correct SCN from PITHASH1 db and modify below. Start replication atcsn

```
start rpspwp01 atcsn 8397003383021
```

Setup GoldenGate replication from Hash RDS (B) to Range RDS (C)

- Shut down application & ensure no application connections
- On Hash GG hub, create extract and pump [to Range RDS (C)]

```
obey ./diroby/create_epsphp02.oby
obey ./diroby/dblogin_s_psphpp02.oby
register extract epsphp02

start epsphp02
info epsphp02

obey ./diroby/create_ppsprp01.oby
```

```
start ppsprp01
info ppsprp01
```

- Verify trails are coming to Range GG hub

```
ls $GG_HOME/dirdat/psphpp02/
```

- Take Snapshot of Production (A)
- Once snapshot completed, Start applications back
- Restore snapshot as new Range RDS (C)

Final step for goldengate replication from Hash db(B) to Range db(C)

- login to Range db (C)

```
SELECT SUPPLEMENTAL_LOG_DATA_MIN, FORCE_LOGGING FROM V$DATABASE;
exec rdsadmin.rdsadmin_util.alter_supplemental_logging('ADD');
exec rdsadmin.rdsadmin_util.force_logging();
exec rdsadmin.rdsadmin_util.switch_logfile;
exec rdsadmin.rdsadmin_util.set_configuration('archivelog retention hours',72);
commit;
```

- Grant permission to GGT and GGS

```
grant create session to GGT;
grant DBA to GGT;
exec rdsadmin.rdsadmin_dbms_goldengate_auth.grant_admin_privilege(grantee=>'GGT',gra
grant exempt access policy to GGT;
GRANT SELECT ANY DICTIONARY TO GGS;
```

- run rds_sequences.sql on Range db (C)

```
host sed '8,25d' $GG_HOME/sequence.sql > $GG_HOME/rds_sequence.sql
@$GG_HOME/rds_sequence.sql ggs
GRANT EXECUTE ON GGS.updateSequence to GGT;
GRANT EXECUTE ON GGS.replicateSequence to GGT;
drop table GGT.CHECKPOINT;
drop table GGT.CHECKPOINT_LOX;
```

- create credentialstore on Range (C) gg hub

```
ALTER CREDENTIALSTORE ADD USER ggs@psprpp01 PASSWORD "WGgRgK7d#(7eX" ALIAS ggsource;
ALTER CREDENTIALSTORE ADD USER ggt@psprpp01 PASSWORD "WGgRgK7d#(7eX" ALIAS ggtarget;
obey ./diroby/dblogin_s.oby
obey ./diroby/dblogin_t.oby
add checkpointtable
```

- add schematrandata

```
obey ./diroby/dblogin_s.oby
add schematrandata PSPADM
```

- On Range GG hub, create replicat

```
obey ./diroby/create_rpsphp02.oby

register replicat rpsphp02 database
```

- On Range GG Hub, start replicat

```
start rpsphp02
```

- On Hash GG hub, Run following commands to flush sequences

```
GGSCI (ggpsppf501.sbg-psp-ppd.a.intuit.com) 5> dblogin useridalias ggsource_hash_pds
Successfully logged into database.

GGSCI (ggpsppf501.sbg-psp-ppd.a.intuit.com as ggs@PPSPHP01) 6> flush sequence pspadm

2023-05-18 01:09:31  INFO    OGG-15311  Successfully flushed 76 sequence(s) pspadm.*
```

- Setup GG monitoring on Range GG hub
- verify sequence bi-directional replication

```
1. On Range (current live database)
SQL> create sequence pspadm.seq_test;

Sequence created.

2. SQL> select SEQUENCE_OWNER, SEQUENCE_NAME, LAST_NUMBER from dba_sequences where S

SEQUENCE_OWNER       SEQUENCE_NAME        LAST_NUMBER
-------------------- -------------------- -----------
PSPADM               SEQ_TEST                       1

3. Test the sequence

a) Run it on Range db(A)
SQL> select pspadm.seq_test.nextval from dual;

   NEXTVAL
----------
         1

b) Wait for 10 secs. Run it on Hash db (B)
SQL> select SEQUENCE_OWNER, SEQUENCE_NAME, LAST_NUMBER from dba_sequences where SEQU

SEQUENCE_OWNER       SEQUENCE_NAME        LAST_NUMBER
-------------------- -------------------- -----------
```

```
PSPADM                    SEQ_TEST                        22

c) Wait for 10 secs. Run it on Range db (C)
SQL> select SEQUENCE_OWNER, SEQUENCE_NAME, LAST_NUMBER from dba_sequences where SEQU

SEQUENCE_OWNER            SEQUENCE_NAME            LAST_NUMBER
-------------------- -------------------- -----------
PSPADM                    SEQ_TEST                        43
```

Prod

```
1. On Range (current Production)
SQL> select SEQUENCE_OWNER, SEQUENCE_NAME, LAST_NUMBER from dba_sequences where SEQU

SEQUENCE_OWNER            SEQUENCE_NAME            LAST_NUMBER
-------------------- -------------------- -----------
PSPADM                    TEST                           777

2. Test sequence

SQL> select pspadm.test.currval from dual

   CURRVAL
----------
       778

b) Wait for 10 secs. Run it on Hash db (B)
SQL> select pspadm.test.nextval from dual;

   NEXTVAL
----------
       798

c) Wait for 10 secs. Run it on Range db (C)
SQL> select pspadm.test.nextval from dual;

   NEXTVAL
----------
       820
```

No labels